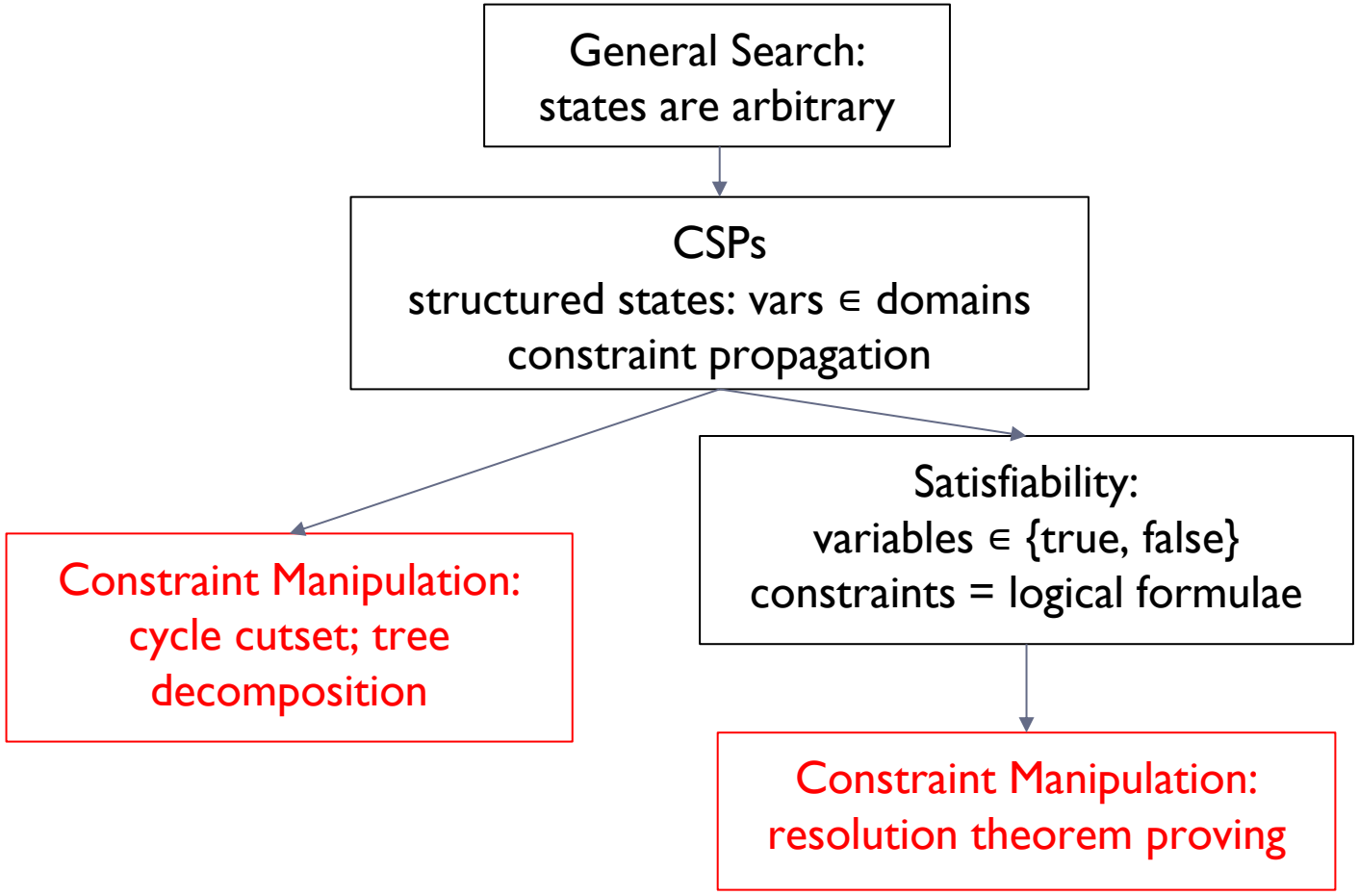
Artificial Intelligence:

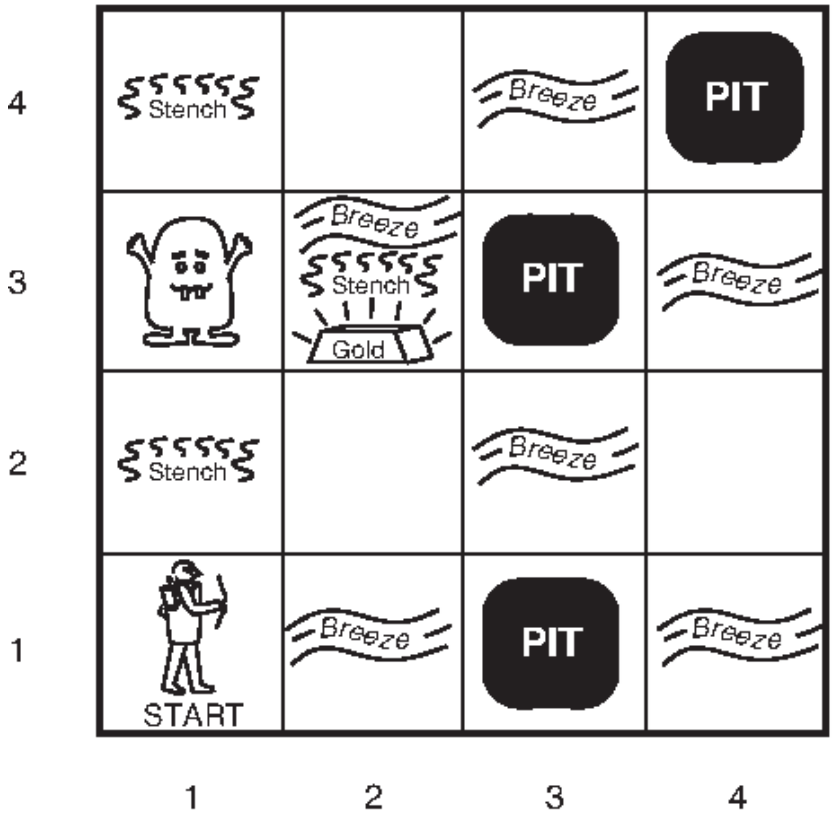
Propositional Logic and Resolution

The Big Picture



Logical Reasoning as CSP

Find variable assignments ("models") that satisfy all constraints.



Wumpus World

B_{ij} = breeze felt

S_{ij} = stench smelt

P_{ij} = pit here

W_{ij} = wumpus here

G = gold

Constraint Language: Syntax

- Sentence: atomic or complex
- Atomic sentence: P, Q, R, …, True, False
- Complex sentence:
  - (Sentence) | [Sentence]
  - ¬ Sentence "negation"
  - Sentence ∧ Sentence "conjunction"
  - Sentence ∨ Sentence "disjunction"
  - Sentence ⇒ Sentence "implies" / "if-then"
  - Sentence ⇔ Sentence "biconditional" / "iff"

Notes on Connectives

$\alpha \vee \beta$ is inclusive or, not exclusive

$\alpha \Rightarrow \beta$ is equivalent to $\neg\alpha \vee \beta$

$\alpha \Leftrightarrow \beta$ is equivalent to $(\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)$

- Prove it!

$\alpha \Rightarrow \beta$ is equivalent to $\neg\alpha \vee \beta$

α	β	$\alpha \Rightarrow \beta$	$\neg\alpha$	$\neg\alpha \vee \beta$
F	F	T	T	T
F	T	T	T	T
T	F	F	F	F
T	T	T	F	T

When two expressions produce the same result in all possible models, their equivalence is a **tautology**.

$\alpha \Leftrightarrow \beta$ is equivalent to $(\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)$

α	β	$\alpha \Leftrightarrow \beta$	$\alpha \Rightarrow \beta$	$\beta \Rightarrow \alpha$	$(\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)$
F	F	T	T	T	T
F	T	F	T	F	F
T	F	F	F	T	F
T	T	T	T	T	T

The equivalence is a **tautology** because it's true in all models. Expressed as a logical sentence:

$$(\alpha \Leftrightarrow \beta) \Leftrightarrow [(\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)]$$

Literals

- Positive literal: P
- Negative literal: ¬ P

- Literals represent simple claims about the world.
- Literal sentences are **unary constraints** on the model.

Literal (Unary Constraint)	Model
P	*P* = True
¬ P	*P* = False

Sentences As Constraints

A sentence with *n* variables functions as an *n*-ary constraint on possible models:

$$[(p \wedge \neg q) \vee (q \wedge \neg p)] \Rightarrow r$$

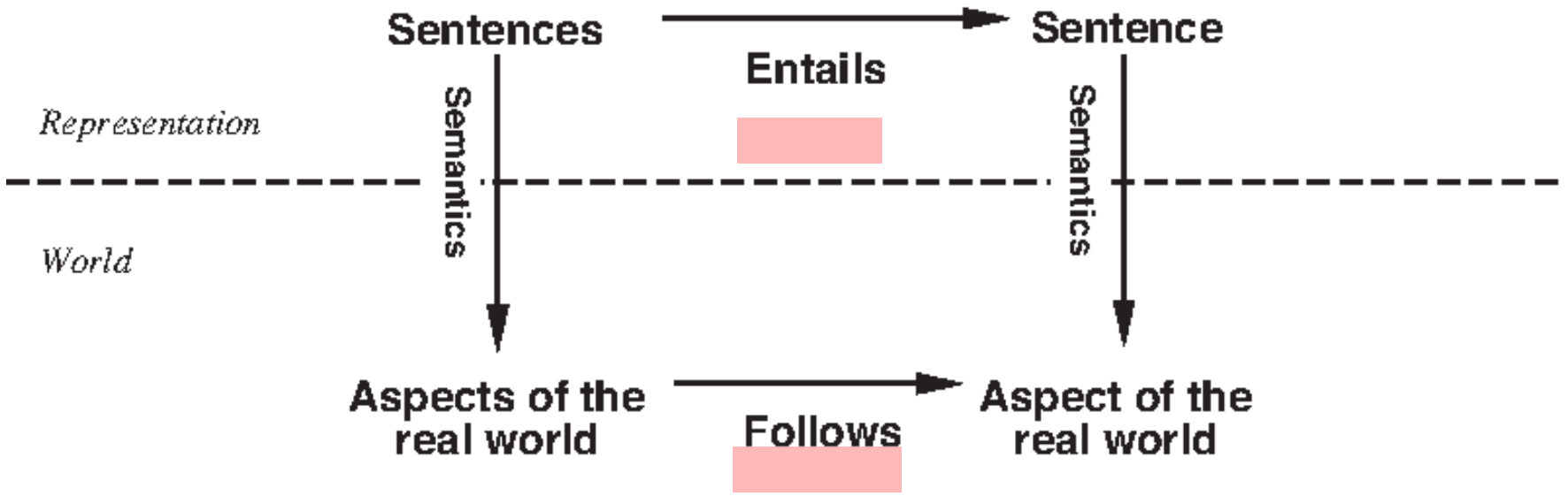
Possible Models

P	Q	R
false	false	false
false	false	true
~~false~~	~~true~~	~~false~~
false	true	true
~~true~~	~~false~~	~~false~~
true	false	true
true	true	false
true	true	true

Semantics: Entailment

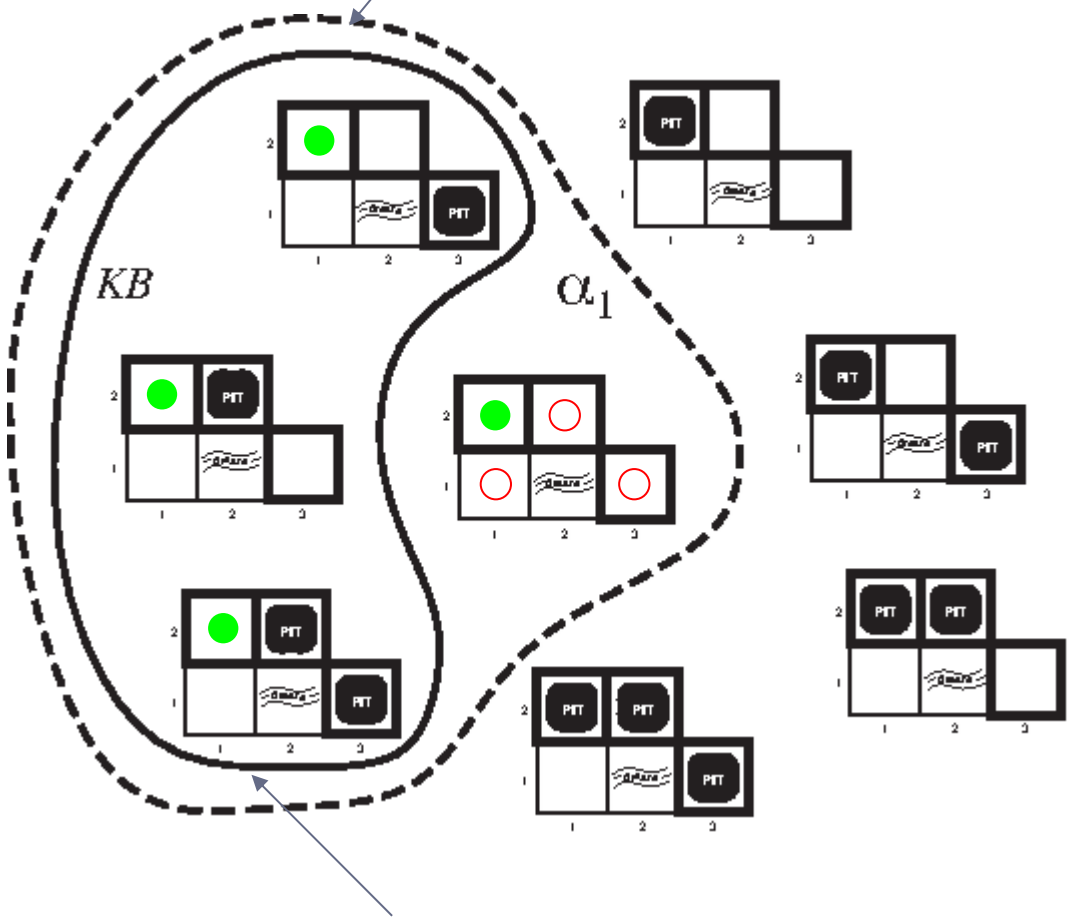
- $\alpha \models \beta$ if and only if, in every model in which α is true, β is also true.
- $M(\alpha)$ denotes the set of all models of α.
- $\alpha \models \beta$ means $M(\alpha) \subseteq M(\beta)$

Semantics: Grounding



Inference By Model Checking

α_1 is models with no pit at 1,2.
$\alpha_1 = M(\neg P_{1,2})$



All models consistent with KB and constraints.
$M(KB) \subseteq M(\alpha_1)$ so $KB \models \alpha_1$. Conclude $\neg P_{1,2}$.

At 1,1: nothing.

$\neg P_{1,1}$, $\neg B_{1,1}$

At 2,1: breeze

$B_{2,1}$, $\neg P_{2,1}$

} KB

Constraints:

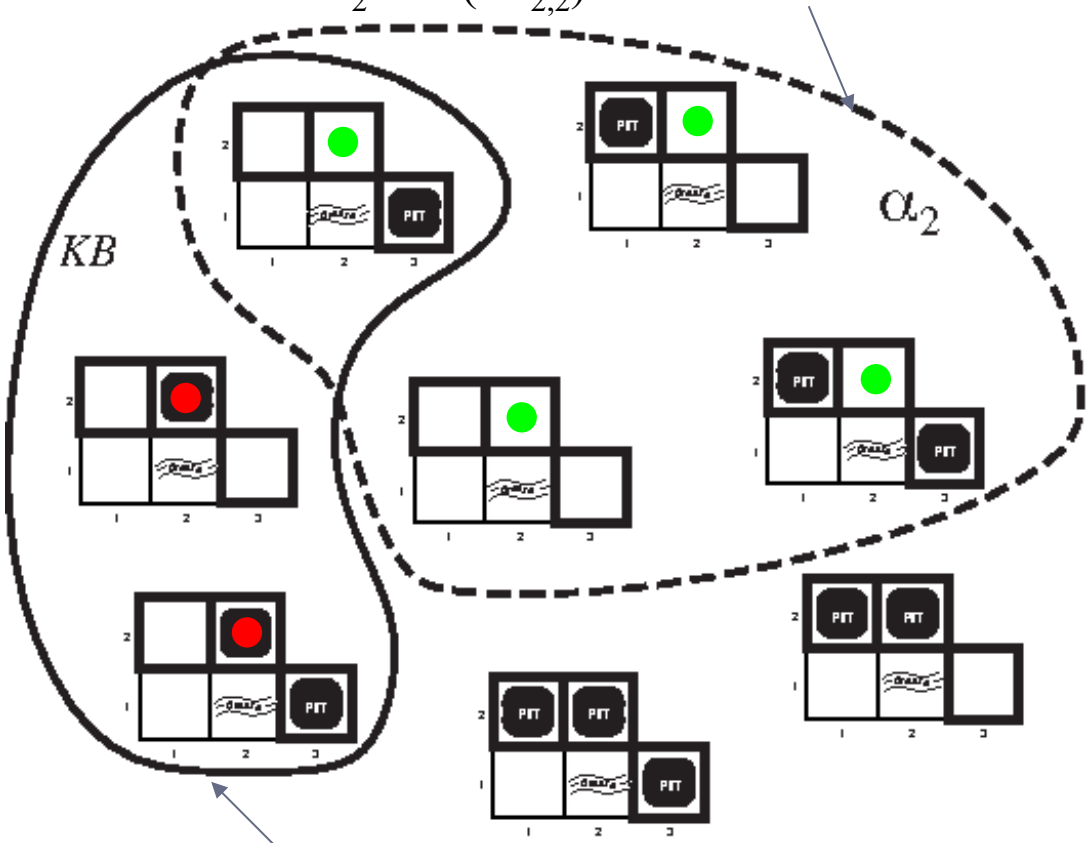
$P_{2,1} \Rightarrow B_{1,1} \wedge B_{2,2} \wedge B_{3,1}$

$B_{2,1} \Leftrightarrow P_{1,1} \vee P_{2,2} \vee P_{3,1}$

Inference By Model Checking

α_2 is models with no pit at 2,2

$\alpha_2 = M(\neg P_{2,2})$



All models consistent with KB and constraints.

$M(KB) \not\subseteq M(\alpha_2)$ so $KB \not\models \alpha_2$. Can't conclude $\neg P_{2,2}$.

At 1,1: nothing.

$\neg P_{1,1}$, $\neg B_{1,1}$

At 2,1: breeze

$B_{2,1}$, $\neg P_{2,1}$

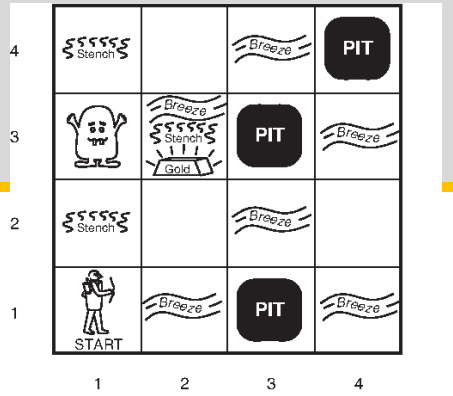
} KB

Constraints:

$P_{2,1} \Rightarrow B_{1,1} \wedge B_{2,2} \wedge B_{3,1}$

$B_{2,1} \Leftrightarrow P_{1,1} \vee P_{2,2} \vee P_{3,1}$

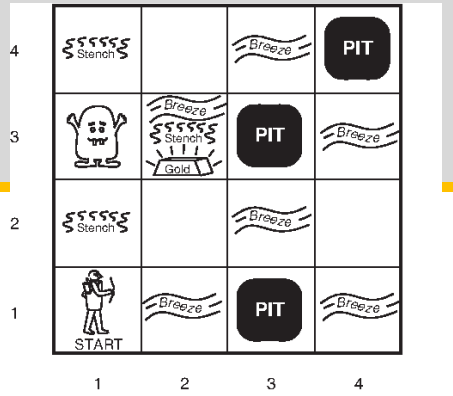
Wumpus World Agent



Step	Agent Action	Tell KB (variable assignment)
1	Start at 1,1. Sense: nothing.	$\neg P_{1,1}$, $\neg B_{1,1}$
	Model check. Relevant constraints: $P_{2,1} \Rightarrow B_{11}$, $P_{1,2} \Rightarrow B_{11}$ Constraint is false in any model with $P_{2,1}$. So KB $\models \neg P_{2,1}$. Similarly, KB $\models \neg P_{1,2}$.	$\neg P_{2,1}$, $\neg P_{1,2}$
2	Agent moves to 2,1. Sense: breeze.	$B_{2,1}$
	Model check. Relevant constraints: $B_{2,1} \models P_{1,1} \vee P_{2,2} \vee P_{3,1}$ There could be a pit at 2,2 or 3,1 or both. Can't infer any new literals.	--



Wumpus World Agent



Step	Agent Action	Tell KB (variable assignment)
3	Agent returns to 1,1.	--
4	Agent moves to 1,2. Sense stench, no breeze.	$\neg B_{1,2}$
	Model check. Relevant constraints: $P_{2,2} \Rightarrow B_{12}$, $P_{1,3} \Rightarrow B_{12}$ Constraint is false in any model with $P_{2,2}$. So KB $\models \neg P_{2,2}$. Similarly, KB $\models \neg P_{1,3}$. Another relevant constraint: $B_{1,2} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$ Constraint propagation gives: $P_{3,1}$.	$\neg P_{2,2}$, $\neg P_{1,3}$ $P_{3,1}$

Cost of Model Checking Inference

- A 4x4 wumpus world has 16 squares.
- Four variables per square: P_{ij}, B_{ij}, S_{ij}, W_{ij}
  - There are 64 variables.

- Search space is 2^{64} possible models!
  - This could take a while…

- What can we do?

Option 1: Exploit Locality

- A constraint only references a square and its four neighbors.
  - Solve subproblems for smaller regions.
  - Share variable assignments across subproblems to get a global solution.

- Constraints on P/B don't mention S or W.
  Constraints on S/W don't mention P or B.
  - Separate the P/B and S/W models.
  - Reduces state space from 2^n to $2 \times 2^{n/2}$.

Option 2: Go Meta

- Instead of searching the space of models, let's search the space of constraints (logical formulae).
- Adopt **inference rules** to **derive** new formulae from old.
  - This is called **theorem proving.**
- Deriving new literals will directly give us valid models.

Inference Rule 1: Modus Ponens

Latin for "mode that affirms".

$$\frac{\alpha \Rightarrow \beta \quad , \quad \alpha}{\beta}$$

Inference Rule 2: And-Elimination

$$\frac{\alpha \wedge \beta}{\alpha}$$

Inference Rule 3: And-Introduction

$$\frac{\alpha \quad , \quad \beta}{\alpha \wedge \beta}$$

Inference Rules Resulting From Logical Equivalences (1/4)

- Commutativity:

$$(\alpha \wedge \beta) \equiv (\beta \wedge \alpha)$$
$$(\alpha \vee \beta) \equiv (\beta \vee \alpha)$$

- Associativity:

$$((\alpha \wedge \beta) \wedge \gamma) \equiv (\alpha \wedge (\beta \wedge \gamma))$$
$$((\alpha \vee \beta) \vee \gamma) \equiv (\alpha \vee (\beta \vee \gamma))$$

Inference Rules Resulting From Logical Equivalences (2/4)

- Double-negation elimination:
  $\neg(\neg\alpha) \equiv \alpha$

- Contraposition:
  $(\alpha \Rightarrow \beta) \equiv (\neg\beta \Rightarrow \neg\alpha)$

Inference Rules Resulting From Logical Equivalences (3/4)

- Implication elimination:

$$(\alpha \Rightarrow \beta) \equiv (\neg\alpha \vee \beta)$$

- Biconditional elimination:

$$(\alpha \Leftrightarrow \beta) \equiv ((\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha))$$

Inference Rules Resulting From Logical Equivalences (4/4)

- De Morgan's Laws:

$$\neg (\alpha \wedge \beta) \equiv (\neg\alpha \vee \neg\beta)$$
$$\neg (\alpha \vee \beta) \equiv (\neg\alpha \wedge \neg\beta)$$

- Distributivity:

$$(\alpha \wedge (\beta \vee \gamma)) \equiv ((\alpha \wedge \beta) \vee (\alpha \wedge \gamma))$$
$$(\alpha \vee (\beta \wedge \gamma)) \equiv ((\alpha \vee \beta) \wedge (\alpha \vee \gamma))$$

Soundness and Completeness

- Entailment: $\alpha \models \beta$ means β is true in all models of α.
- Derivation: $\alpha \vdash_I \beta$ means β can be derived from α using the inference rules in I.

- **Soundness of I**: if $\alpha \vdash_I \beta$ then $\alpha \models \beta$
  - Using I, we only derive entailed sentences.
  - "We don't make stuff up."

- **Completeness of I:** if $\alpha \models \beta$ then $\alpha \vdash_I \beta$
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  - "If it's true, we can deduce it."

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Validity and Satisfiability

- A sentence α is **valid** if it is true in all models. We denote this by $\models\alpha$.
- A sentence α is **satisfiable** if it is true in at least one model.
- If a sentence α is valid, its negation $\neg\alpha$ is not satisfiable.
  - To prove α, show that $\neg\alpha$ is unsatisfiable.

Theorem Proving

- Start with axioms and givens, and apply inference rules to derive new formulae.
- But there are $O(2^N)$ formulae in N vars!
- And most are useless.
- How can we focus on formulae relevant to proving our goal?

Resolution Theorem Proving

- The good news: there is a single inference rule that is sound, complete, and can find proofs with reasonable efficiency.
- The bad news: it only works for problems in **conjunctive normal form.**
- But any sentence can be rewritten in conjunctive normal form.
- … but length may increase exponentially.

Resolution Theorem Proving

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- But any sentence can be rewritten in conjunctive normal form.
- … but length may increase exponentially.

Definition: Clause

A **clause** is a disjunction of literals:

$$\ell_1 \vee \ell_2 \vee \ldots \vee \ell_n$$

The ℓ_i may be positive or negative.

A single literal ℓ is called a **unit clause.**

Definition: Conjunctive Normal Form

A sentence is in **conjunctive normal form** if it is a conjunction of clauses:

$$c_1 \wedge c_2 \wedge \ldots \wedge c_m$$

Example:

$$(P \vee \neg Q \vee R) \wedge (Q \vee S) \wedge (P \vee \neg R \vee T)$$

Converting to CNF

$B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$	Given
$(B_{1,1} \Rightarrow (P_{1,2} \vee P_{2,1})) \wedge$ $((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1})$	Biconditional elimination
$(\neg B_{1,1} \vee P_{1,2} \vee P_{2,1}) \wedge$ $(\neg(P_{1,2} \vee P_{2,1}) \vee B_{1,1})$	Implication elimination (twice)
$(\neg B_{1,1} \vee P_{1,2} \vee P_{2,1}) \wedge$ $((\neg P_{1,2} \wedge \neg P_{2,1}) \vee B_{1,1})$	De Morgan
$[\ (\neg B_{1,1} \vee P_{1,2} \vee P_{2,1}) \wedge$ $(\neg P_{1,2} \vee B_{1,1}\ ) \wedge$ $(\neg P_{2,1} \vee B_{1,1})\ ]$	Distributivity Result is in CNF.

Converting to CNF

$$(a \wedge b) \vee (c \wedge d) \vee (e \wedge f)$$

This sentence happens to be in DNF.

converts to

$$(a \vee c \vee e) \wedge (a \vee c \vee f) \wedge$$
$$(a \vee d \vee e) \wedge (a \vee d \vee f) \wedge$$
$$(b \vee c \vee e) \wedge (b \vee c \vee f) \wedge$$
$$(b \vee d \vee e) \wedge (b \vee d \vee f) \wedge$$

The Resolution Inference Rule

$$\frac{\ell_1 \vee \ell_2 \vee \ldots \vee \ell_k \quad , \quad m_1 \vee m_2 \vee \ldots \vee m_n}{\ell_1 \vee \ldots \vee \ell_{i-1} \vee \ell_{i+1} \vee \ldots \vee \ell_k \vee m_1 \vee \ldots \vee m_{j-1} \vee m_{j+1} \vee \ldots \vee m_n}$$

where ℓ_i and m_j are complementary literals,
i.e., ℓ_i is $\neg m_j$.

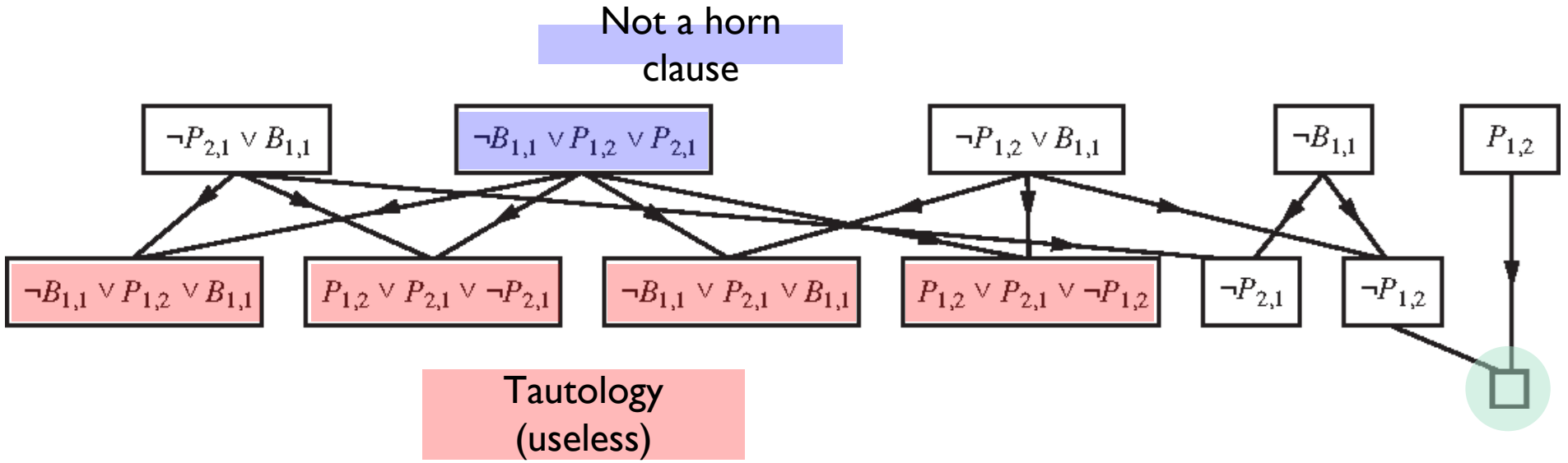
From two clauses of length *k* and *n*, derives a new clause of length *k+n-2*.

Contradictions Generate An Empty Clause

- $P \wedge \neg P$ is a **contradiction**, and therefore **unsatisfiable**.
- Resolving P with ¬P yields an empty clause.

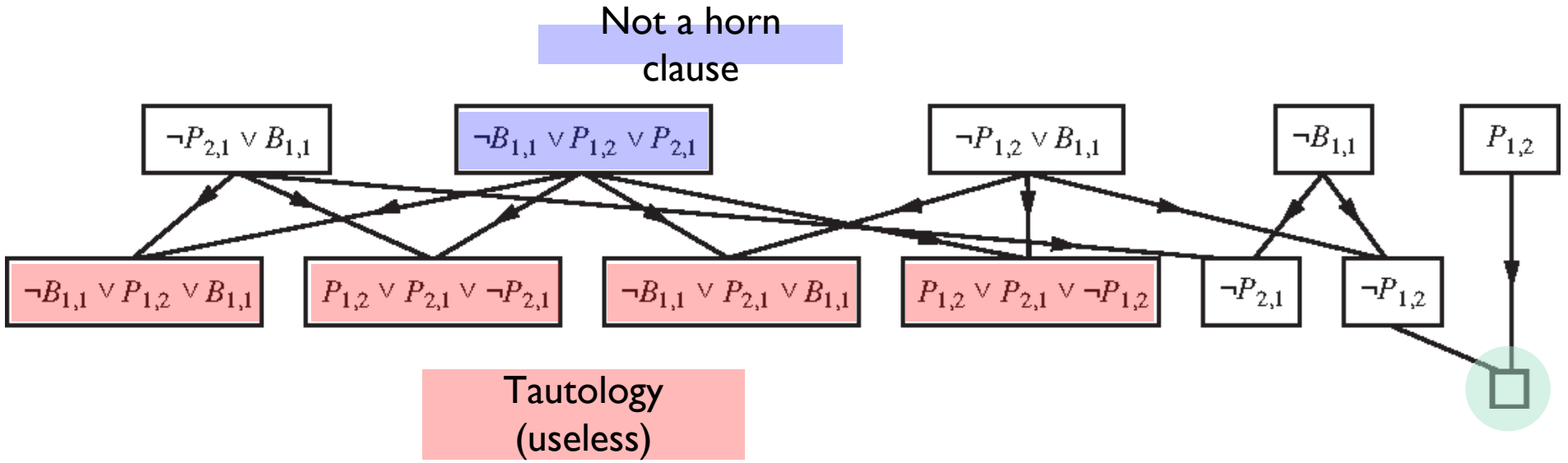
$$\frac{P \quad , \quad \neg P}{\square}$$

Wumpus Reasoning by Resolution



To prove $\neg P_{1,2}$, add $P_{1,2}$ to the set of givens and use resolution to derive a contradiction.

Wumpus Reasoning by Resolution



To prove $\neg P_{1,2}$, add $P_{1,2}$ to the set of givens and use resolution to derive a contradiction.

More Efficient Inference

- Resolution is complete, but even with only one inference rule, the worst case cost is exponential in the # of symbols.
- But for problems that can be formulated using **Horn clauses**, inference can be more efficient.
- With **definite clauses** entailment can be decided in *linear* time.

Summary

- Propositional logic is a constraint language.
- Model checking can be tractable if you can exploit locality in the state space.
- Propositional entailment is co-NP Complete, so worst case is $O(2^N)$.
- In practice, many problems can be solved efficiently by resolution, especially if they can be formulated using definite clauses.